📘 SESSION 12 – DECISION TREE (CLASSIFICATION)

🎯 Topic

- What is Decision Tree
- Tree structure (root, nodes, leaves)
- Splitting logic
- Gini Index & Entropy (intuition only)
- Overfitting concept
- Practical implementation

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1. 🎯 Learning Objectives

By the end of this session, students will be able to:

- Understand how **Decision Trees work**
- Learn tree components (root, nodes, leaves)
- Understand how data is **split**
- Learn concept of **Gini / Entropy (basic intuition)**
- Build a **Decision Tree classifier**
- Understand **overfitting problem**

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2. 🔁 Quick Recap (Connect Previous Session)

Start class like this:

"Till now we used Logistic Regression and Naive Bayes.
These models are mathematical.
Today we will learn a model that works like human decision-making — Decision Tree."

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3. 📖 What is Decision Tree?

3.1 Definition

A Decision Tree is a supervised learning algorithm that makes decisions using a **tree-like structure**.

3.2 Simple Meaning (Most Important Line)

"Decision Tree splits data step-by-step based on conditions to make a decision."

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4. 🌳 Structure of Decision Tree

Components

🔹 **Root Node**

- Starting point
- First decision

🔹 **Internal Nodes**

- Decision points
- Conditions (e.g., Study Hours > 5)

🔹 **Leaf Nodes**

- Final output
- Class label

Teaching Line

"Decision Tree works like a flowchart of decisions."

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5. 📊 Example (Very Important)

Student Pass/Fail

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Is Study Hours > 4?
  ├── Yes → Pass
  └── No  → Fail
```

👉 Model learns simple rules like this.

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6. 📊 How Decision Tree Works

Step-by-Step

1. Select best feature
2. Split dataset
3. Repeat splitting
4. Stop when:
    - Data is pure OR
    - Depth limit reached

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7. 📊 How Tree Decides Best Split?

Concept: Purity

- Pure node → all same class
- Impure node → mixed classes

Goal

👉 Increase purity after each split

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8. 📊 Gini Index (Concept Only)

Meaning

Measures impurity of data

Range

- 0 → pure
- Higher → more mixed

Teaching Line

"Lower Gini = better split"

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9. 📊 Entropy (Concept Only)

Meaning

Measures disorder/randomness

Teaching Line

"Lower entropy = better decision"

👉 Don't go into formulas, only intuition.

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10. 📊 Advantages

✅ Easy to understand
✅ No need for scaling
✅ Handles non-linear data
✅ Visual representation

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11. ⚠️ Overfitting Problem

What happens?

Tree becomes too complex:

- Memorizes data
- Performs poorly on new data

Example

- Very deep tree
- Too many splits

Solution

- Limit depth
- Pruning

Teaching Line

"A very deep tree is not always a good tree."

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12. 🎯 Interview Concepts

Q1: What is Decision Tree?

👉 Tree-based model for decision making.

Q2: What is root node?

👉 First split of the tree.

Q3: What is Gini Index?

👉 Measure of impurity.

Q4: What is overfitting?

👉 Model memorizes data instead of learning.

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13. 💻 PRACTICAL (YOUR FORMAT)

🎯 **Objective**

Predict Pass/Fail using Decision Tree

14. 📊 Output Interpretation

Explain:

- Tree shows decision rules
- Model creates conditions automatically
- Easy to understand decisions

👉 Example Rule:

"If Study Hours > 4 → Pass"

"Decision Trees are powerful because they mimic human decision-making.
They are easy to understand, visualize, and apply, but must be controlled to avoid overfitting."